Bleep

PROGRAMMING PRACTICE QUESTIONS

Learn by doing.

This practice set contains *50 Python programming questions* covering Python Basics, String Slicing, Lists, List Comprehension, Loops, Functions, Recursion, and Try & Except.

The objective is to move from understanding Python concepts to applying them through code.

Work through all the questions. Write, run, test, and improve your solutions.

LEVEL 1 — INTERMEDIATE BEGINNER

20 Questions

01. Marks, Percentage & Grade

Write a program to input marks in five subjects and calculate the total, percentage, and grade.

02. Leap Year

Write a program to determine whether a given year is a leap year.

03. Largest & Second Largest

Write a program to find the largest and second largest among three numbers.

04. Electricity Bill

Write a program to calculate an electricity bill based on slab rates.

05. Armstrong Number

Write a program to check whether a number is an Armstrong number.

06. String Slicing

Input a string and print the first 3 characters, last 3 characters, every alternate character, and the reversed string.

07. Character Classification

Count the number of vowels, consonants, digits, and special characters in a string.

08. Palindrome Using Slicing

Check whether a string is a palindrome using slicing.

09. Character Frequency

Find the frequency of each character in a string.

10. Spaces & Word Count

Remove all spaces from a string and count the number of words.

11. Prime Numbers

Print all prime numbers between 1 and N.

12. Fibonacci Sequence

Generate the Fibonacci sequence up to N terms using loops.

13. Factors of a Number

Find all factors of a given number.

14. Perfect Number

Check whether a number is a Perfect Number.

15. Floyd's Triangle

Print Floyd's Triangle up to N rows.

16. List Statistics

Find the largest, smallest, and average value in a list.

17. Remove Duplicates

Remove duplicate elements from a list while maintaining the original order.

18. Second Largest & Second Smallest

Find the second largest and second smallest element in a list.

19. Even & Odd Lists

Separate even and odd numbers from a list into two different lists.

20. List Rotation

Rotate a list by K positions to the left.

LEVEL 2 — INTERMEDIATE

20 Questions

01. Squares Using List Comprehension

Create a list of squares of numbers from 1 to 50 using list comprehension.

02. Prime Numbers Using List Comprehension

Generate a list containing only prime numbers between 1 and 100 using list comprehension.

03. Convert Strings to Uppercase

Convert a list of strings into uppercase using list comprehension.

04. Word Lengths

Create a list containing the lengths of each word in a sentence.

05. Remove Empty Strings

Remove empty strings and strings containing only spaces from a list using list comprehension.

06. Prime Function

Write a function to check whether a number is prime.

07. GCD Function

Write a function to calculate the GCD of two numbers.

08. Second Largest Function

Write a function that returns the second largest element of a list.

09. Longest Word

Write a function that accepts a sentence and returns the longest word.

10. Merge & Remove Duplicates

Write a function that merges two lists and removes duplicate elements.

11. Common Elements

Find the common elements between two lists without using sets.

12. Missing Number

Find the missing number from a list containing numbers from 1 to N.

13. Element Frequency

Find the frequency of every element in a list.

14. Sort Without sort()

Sort a list in ascending order without using the sort() method.

15. Diamond Pattern

Print a diamond star pattern.

16. Safe Division

Write a program to safely divide two numbers using exception handling.

17. Valid Integer Input

Accept integer input until a valid integer is entered using try-except.

18. Handle IndexError

Handle IndexError while accessing list elements.

19. Handle FileNotFoundError

Handle FileNotFoundError while opening a file.

20. Menu-Driven Calculator

Build a menu-driven calculator that handles invalid inputs and division by zero using try-except.

LEVEL 3 — HARD

10 Questions

01. Factorial Using Recursion

Find the factorial of a number using recursion.

02. Fibonacci Using Recursion

Generate the Fibonacci sequence using recursion.

03. Sum of Digits Using Recursion

Find the sum of digits of a number using recursion.

04. Reverse a String Using Recursion

Reverse a string using recursion.

05. Tower of Hanoi

Solve the Tower of Hanoi problem using recursion.

06. Recursive Palindrome

Write a recursive function to determine whether a string is a palindrome.

07. Prime Numbers Using List Comprehension & Helper Function

Use list comprehension and a helper function to create a list containing only prime numbers.

08. Numbers Until Stop

Read numbers until stop is entered, handle invalid inputs using try-except, then calculate minimum, maximum, average, and median.

09. Flatten a Nested List

Write a recursive function to flatten a nested list.

10. Student Record Management System

Create a Student Record Management System using functions, lists, loops, list comprehensions, exception handling, and file handling.

HOW TO WORK

Do not treat these as questions to read and forget.

For every question:

Read → Think → Code → Run → Test → Fix → Understand

Write your own solution.

Test different inputs.

Look for errors.

Fix them.

Then move forward.

The objective is not simply to complete 50 questions. The objective is to become more capable of solving problems independently.

Learn by doing.